Topic: Second Shifting Theorem

Question: For $x \geq 0$, rewrite the function with a 2-unit shift to the right.

$$f(x) = e^{3x} - 5x^2 + 7x \sin(4x)$$

Answer choices:

A $(e^{3x} - 5x^2 + 7x \sin(4x))u(x-2)$ $x \geq 0$

B $(e^{3(x-2)} - 5(x-2)^2 + 7(x-2)\sin(4(x-2)))u(x-2)$ $x \geq 0$

C $(e^{3(x-2)} - 5(x-2)^2 + 7(x-2)\sin(4(x-2)))u(x-2)$ $x \geq 2$

D $e^{3(x-2)} - 5(x-2)^2 + 7(x-2)\sin(4(x-2))$ $x \geq 2$

© Krista King Math

Solution: B

Because we want to shift the function 2 units to the right, we can start by substituting $c = 2$ into our formula.

$$f(x-c)u(x-c) = \begin{cases} 0 & 0 \leq x < c \\ f(x-c) & x \geq c \end{cases}$$

$$f(x-2)u(x-2) = \begin{cases} 0 & 0 \leq x < 2 \\ f(x-2) & x \geq 2 \end{cases}$$

Now we just need to replace $f(x-2)$, which is

$$f(x) = e^{3x} - 5x^2 + 7x\sin(4x)$$

$$f(x-2) = e^{3(x-2)} - 5(x-2)^2 + 7(x-2)\sin(4(x-2))$$

So the shifted function, for $x \geq 0$, is represented by

$$(e^{3(x-2)} - 5(x-2)^2 + 7(x-2)\sin(4(x-2)))u(x-2)$$

© Krista King Math



Topic: Second Shifting Theorem

Question: What is the Laplace transform of the function?

$$f(x) = \begin{cases} 0 & 0 \leq t < 5 \\ (x-5)\sin(x-5) & t \geq 5 \end{cases}$$

Answer choices:

A $\mathscr{L}(f(x)) = e^{5s}\left(\dfrac{2s}{(s^2+1)^2}\right)$

B $\mathscr{L}(f(x)) = \dfrac{2s}{(s^2+1)^2}$

C $\mathscr{L}(f(x)) = e^{-5s}\left(\dfrac{2s}{(s^2+1)^2}\right)$

D $\mathscr{L}(f(x)) = e^{s}\left(\dfrac{2s}{(s^2+1)^2}\right)$

© Krista King Math

Solution: C

The function $f(x)$ is a shifted version of the function $g(x) = x \sin x$ for $x \geq 0$ by 5 units to the right. In other words $f(x) = g(x-5)u(x-5)$,

$$\mathscr{L}(g(t-c)u(t-c)) = e^{-cs}G(s)$$

$$\mathscr{L}(f(x)) = \mathscr{L}(g(x-5)u(x-5)) = e^{-5s}G(s)$$

where $G(x)$ is the Laplace transform of the function $g(x)$.

$$G(s) = \mathscr{L}(g(x)) = \mathscr{L}(x \sin x)$$

Using the transform formula,

$$\mathscr{L}(t \sin(at)) = \frac{2as}{(s^2+a^2)^2}$$

we get

$$G(s) = \mathscr{L}(g(x)) = \frac{2s}{(s^2+1)^2}$$

Then the transform of $f(x)$ is

$$\mathscr{L}(f(x)) = e^{-5s}\left(\frac{2s}{(s^2+1)^2}\right)$$

© Krista King Math

Topic: Second Shifting Theorem

Question: Which function represents $f(t)$, shifted 7 units to the right, for $t \geq 0$?

Answer choices:

A $f(t-7)u(t-7)$

B $f(t)u(t-7)$

C $f(t-7)u(t)$

D $f(t)u(t)$

© Krista King Math



Solution: A

Because we want to shift the function 7 units to the right, we can start by substituting $c = 7$ into the Second Shifting Theorem formula.

$$f(x-c)u(x-c) = \begin{cases} 0 & 0 \leq x < c \\ f(x-c) & x \geq c \end{cases}$$

$$f(x-7)u(x-7) = \begin{cases} 0 & 0 \leq x < 7 \\ f(x-7) & x \geq 7 \end{cases}$$

© Krista King Math